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PROJECT 1 - INITIAL SYSTEM PROPOSAL

SYSTEM: HASTA CAR RENTAL SYSTEM

Group Name: SpinTheWheel

Members:

- 1) Chun Yao Ting (A24CS0239) [Team Leader]**
- 2) Evelyn Ang (A24CS0068)**
- 3) Mirza As-Siddiq Bin Tohari (A24CS0112)**
- 4) Muhammad Aliff Bin Helmi (A24CS0272)**

Stakeholder: Hasta Travel & Tours Sdn. Bhd.

Representative Name	Date Interviewed
Encik Alif	30/10/2025

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1.1 Introduction

Efficient data management is a must for modern information systems, especially in car rental services which are heavily relying on transaction of data. A well-designed database system can allow a company to store, process and retrieve large volumes of operational data accurately, consistently, easily, and in real time as well. Without a database-driven solution, a business will face issues such as data redundancy, data inaccuracy and inefficient workflow.

This project mainly focuses on the analysis and design of a database system for the car rental service operated by Hasta Travel & Tours Sdn. Bhd. As for now, the operation relies heavily on manual methods, such as WhatsApp, handwritten records, Excel records, physical whiteboards and Wahdah staff system. Hence, we can see the operation is very fragmented, as customer information, bookings, payments and vehicle availability are all managed in different places. The fragmented data storage mode causes issues such as duplicated data entry, difficult to track transactions, and heavily dependent on staff work.

The objective of this report is to identify the database requirements of the current system and propose a centralized database to support real-time data transaction, data integrity and fast data retrieval. The proposed system must systematically store and manage the data of entities, such as users, customers, vehicles, bookings, and payments. Meanwhile, it also needs to align with the business rules and relationship between entities.

The scope of this project is the database analysis and design phase of the car rental system. This includes the study of current transaction processes, identifying the data requirements, defining entities, attributes, and relationships, and specifying the transaction type like data entry, data updates, data delete, and data queries. This project should also cover the clear understanding of the business rules and the system boundaries, such as the major user views, internal system, and external system. Besides, a proposal draft is also given to be reviewed and used in the design phase if approved.

In summary, this report will act as a strong foundation and reference for us to develop the conceptual, logical, and physical database later on.

1.2 Background Study

Hasta Travel is a company operating in Universiti Teknologi Malaysia (UTM), mainly providing their car rental service for UTM students and staff. Hasta Travel provides a highly customizable service for their customers such as delivery service for out-station pickup and return, booking outside working hours, short period booking within one day, cancel booking before one day of pickup and so on. Due to their small customer base, the company staff are still able to manage the operation of the company that highly relies on the manual processes and human judgment on eligibility verification, office-hour constraint and location charges. Although this manual approach allows flexibility and still can fulfil the current requirements, this operation mode will fall off once the booking volume increases as the company grows over the time. Additionally, Hasta also decided to transition from physical operation to mixed operation, where customers can do their booking offline or online. This means that the current operation mode already cannot address the future digitalized booking mode.

Due to the absence of a centralized database, it is very difficult for staff to maintain the accuracy of data and make sure the data is up to date. For example, booking availability is checked manually, which is difficult to ensure the vehicle booked is not double booked under overlapping booking periods. Besides, payment verification, booking approval and vehicle status updates are also handled independently, which cause delays and inconsistency across records. In fact, the real practice is the data transaction should reflect changes on other related data. Hence, we can observe that there is a critical issue of a lack of real-time data processing, which is disastrous for car rental services that are sensitive to time.

Other than that, the current system also stores customer documents, such as driving licenses, matric cards, identification cards, inspection form, agreement forms and additional documents in different formats and different systems. The storage of data is unorganized and unstructured, and this causes the historical customer and booking records to be very hard to retrieve in future. The lack of proper data integrity constraints also might cause invalid or incomplete data to be recorded. Additionally, staff also need to manually compile booking statistics, revenue summaries, and vehicle usage records, which is inefficient and increases the rate of human errors. Without a relational database, data cannot be easily queried to support decision making.

Overall, we can see there are many limitations in current business operation mode. Therefore, we need to implement a centralized relational database to store customer, vehicle, booking and transaction data in a systematic way, with clearly defined relationships between data and entities. This allows real-time updates in data, as well as reduces data redundancy by normalization, and also ensures data consistency and integrity for all operations. With this proposed solution on database, efficient querying of data and transaction handling can be fulfilled, which can reduce staff workload and overall service quality for customers.

1.3 Problem Statement

Based on the case study, we can observe that Hasta Travel still uses a fully manual and fragmented workflow in the company operation, such as the process of booking, pickup, return and customer management. This had caused a heavy workload on the staff side, resulting in operational efficiency, and increasing risk of human error. Meanwhile, for the customer side, the booking, pickup and return process is also inefficient, which affects the customer experience and brings a negative effect on company image.

Currently, the booking process fully relies on WhatsApp communication, which requires staff to repeatedly prompt customers to send information, manually verify their eligibility to book, check booking constraints such as location, working hours and minimum rental period, as well as determine additional charges for different types of special cases. Meanwhile, customers also have to manually send the details, and their details are not stored, meaning that they still need to send the send information again during the next booking, which is unnecessary and complicated for the customer booking process.

Besides, staff even need to check the car availability to approve the booking by using a physical whiteboard and a separate staff system which is Wahdah. They also need to manually record down booking details across different platforms, including WhatsApp, Wahdah, physical whiteboards, notebook, and Excel daily sales report. This is a bad practice for data transactions, as the data redundancy, delays and errors are likely to occur.

Furthermore, staff also must manually generate, print, fill, verify and store inspection and agreement forms signed by customers during pickup. Important information such as mileage, fuel level, pickup time and staff in charge are required to be filled in by staff manually, as these details are also not automatically inserted. The document is also stored in an unorganized way, as some of them will be in digital documents, some of them will be manual, so it is hard to track the legal documents if needed. Both online and physical pickups require human intervention, like document handling and updates to sales records repeatedly.

Additionally, staff also depend on external GPS tracking systems as well as manual inspection to record mileage, fuel level, late return penalties, and damages. Any issues during rental such as delays or damages will require documentation, customer communication and penalty calculation, all by manual work from staff. Other than that, deposit refunds are also tracked manually, and return information also needs to be recorded into Excel. This provides no flexibility to customer and staff, and also contributes to repetitive workload for staff.

Overall, the absence of integrated and automated booking and management system causes excessive manual data entry across different platforms, high dependency on staff accuracy, judgment and availability, long processing time for booking and transactions, high risk for data inconsistency, lost records and human error, poor customer service due to long response time and repeated information request, and no scalability to address future growth in company size and booking volume. As Hasta also wishes to have a more flexible workflow by transiting to off-station booking, this requires a full revolution in the current system.

1.4 Proposed Solution

To improve the current system workflow, we came out with our proposed solution which digitalizes the car rental system and organizes the data systematically in the database. We proposed the system to have both the customer side and staff side. The customer side is where the customers do the booking, while the staff side is for the staff to access and manage all the bookings. Next, we will explain the overall workflow of our proposed system.

First, as currently customers need to fill the information again and staff need to record the information again during each booking, we propose a user management system where staff and customer information will be stored in their respective accounts. This will be covered in the authentication module which include register, login, activation, and access control. Staff and customer will use the same login page but will be redirected to staff side or customer side based on their respective role.

To register, the customer has to fill in their full name as per NRIC, email, and their password, and each email should be unique, meaning that email cannot be reused for different accounts. After registration, customers can login into their account by entering correct credentials, but they still cannot do booking because their account is not activated yet. To activate an account, customers have to fill in all the information required such as IC/passport, nationality, UTM ID (Staff/Student), contact, emergency contact, emergency contact relationship, license number, license expiry date, college, college block, faculty, and home address. Afterwards, they also have to submit a front & back picture of IC/passport, UTM matric card, and car license. The submitted information will be sent to the staff side for manual verification. If staff notice any issue, they can decide to reject it with a reason, else they can approve the activation and customers can start booking now. For activation requests that are rejected, customers may resubmit the activation request and wait for staff approval. Additionally, the system will also send email to notify successful or unsuccessful activation.

Now, we propose the digitalized booking process, which is much easier and more convenient than the current booking process. First of all, customers need to select the preferred date, time, and location for both pickup and return. The system will filter based on the time range of booking and display available cars during that period, with the car model, car plate, features and price. Then, customers can select the preferred car and proceed to the payment page. The payment page will have the payment breakdown section to display all the costs such as rental cost, deposit of RM50, and delivery fee (if the pickup location and return location is not in UTM). Customers also could use vouchers to gain discounts and the discounted price will be shown as well. As the stakeholder does not want to implement the online payment gateway, hence payment screenshot is still used for payment submission. Customers can either pay by QR or using Hasta bank details shown in the payment page. Last but not least, they need to submit the receipt or payment screenshot for verification purposes.

Currently, the booking is under pending status. So, staff will review the customer booking and check if the payment receipt is correct or not, and double check the availability of the car. If there is no issue, staff can approve the booking, else they can reject the booking by stating the reasonings and issue the rejection refund by submitting a refund receipt. Customers will also receive email for booking approval and rejection. If customers wish to cancel their booking, they can do so before 24 hours of pickup time. Then, staff need to review it and issue the cancellation refund by submitting a refund receipt.

After booking approval, customers can do pickup before 30 minutes of pickup time, and there will also be email reminders to notify customers to pick up their car. There will be a few steps to finish the pickup process. First, customers have to review the booking details. Secondly, customers need to download the inspection form and agreement form, fill it, and submit the filled form in the system. Thirdly, customers need to take 5 pictures, which are front, back, left, right and odometer of the car and submit it to the system. Finally, customers can finish the pickup by clicking submit, and the rental status will become picked up.

To return the car, customers can issue a return before 24 hours of return time, and customers will receive an email reminder as well. There will be some steps to complete the return as well. First, customers need to review the booking details. Secondly, customers have to submit 6 pictures, which are front, back, left, right, odometer and car key of the car. Thirdly, customers have to submit their feedback, if there is any issue during booking, customers can press the tickbox in the page as well. Fourthly, customers need to fill in their bank details which are cardholder name, bank name, and account number for deposit refund later on. They may use an existing bank detail if they stored the information in their profile. Finally, customers can submit the return form, and the rental status will change to returned.

After the car returns, there are few tasks that need to be done by the staff. First, staff need to check the feedback of customers for this booking. If there is any issue, the car will automatically become maintenance status. Only after the staff solve the issue (e.g. Aircond no function), staff can click “Resolve” and the car will be available. Secondly, staff need to update the mileage and fuel bar of the cars based on the odometer submitted by the customer. Finally, the booking is finally completed after staff issued the deposit refund by submitting the refund receipt.

For late return, the status of booking on the customer side will be changed to late return after 30 minutes of original return time. The penalty imposed will be based on the return time, for every hour after 30 minutes of original return time will be charged for RM50. Customers have to complete the return process first. Afterwards, their account will be directly blacklisted and there will be one pop out to ask the customer to pay the penalty first. Customers cannot do any new booking as their account is suspended. After they submit the payment receipt for penalty, they still need to wait for approval and their account remains suspended. So, the staff will check the payment receipt submitted by the customer and approve it if there is no issue. After that, customers will remove from the blacklist and can do booking as normal.

Now, we will explain our proposed solution to manage vouchers. There will be two methods of gaining vouchers, one generated automatically by system and another one by staff doing voucher assignments to customers. For automatically generated vouchers, it can be generated in two ways, the first method is new customers can receive one NEWUSER10 discount, and the second method is claimed by loyalty cards. For loyalty card features, the customer will receive one loyalty stamp after return of each booking if the booking exceeds 12 hours. After collecting three stamps, they may claim the LOYALTY10 vouchers, which they can enjoy the discount on their future booking. For manual vouchers assignment, this only happens if there is any issue during booking and rewards for loyal customers. So, staff can also assign the vouchers to their account. On the staff side, they are also able to view the voucher's usage rate and history.

Furthermore, there is also car fleet management on the staff side. Staff is able to view the car details in the car list tab, such as the car plate, car model, price rate, features, last booked time and its status. Staff can directly manipulate the status using a dropdown, and also be able to edit the car details such as car picture and price. Staff is able to add new cars as well if there are any new leases in future. They will need to fill in the car plate, car model, car picture, price rate for 1 hours, 3 hours, 5 hours, 7 hours, 9 hours, 12 hours, and 24 hours. Staff are also able to view the past rental records of the cars for tracing purposes.

Additionally, there is customer management on the staff side. Staff can see all the details of each customer, such as their college, faculty, contact number, email, and status, and also can search for certain customers by input substring. The status of the customer also can be changed by dropdown, such as new, active, blacklisted, inactive. The customer is managed using soft delete instead of hard delete by setting the customer status to inactive so that our stakeholder can trace old customer information.

Moreover, there is rental management on the staff side. Staff can view all the rental details such as status, customer, car plate and booking period. Staff also can filter searching by customer name, status, car plate, month, or year. Meanwhile, admins can generate a report of rental by the filter as well. There is also a list of locations offered for pickup and return, and staff can view the delivery price for each location and be able to edit the price as well. Staff can also add a new location by filling in the location name and its delivery price.

Other than that, there is also payment management on the staff side. Staff can view all the transactions done by the customer, with the amount, rental duration and the receipt submitted. Certain transactions also can be filtered based on the customer's name, rental ID, year, or month. Meanwhile, admins can also generate a transaction report by the filter as well.

Now we move to the admin side. Admins have the privilege to view all staff and add new staff by inserting their information such as full name, email, position and password. Admin also have the privilege to view analytics dashboard which show the graph of booking, revenue, customer gain, customer distribution by faculty and college and booking heatmap, filter by last 7 days, last 12 months and last 5 years.

1.4.1 Proposed System Flowchart

This section presents the proposed system workflows through flowcharts. An overall system flowchart gives a comprehensive view of the entire process flow, while six sub-flowcharts detail specific processes such as Activation, Booking, Pickup, Return, Staff Issue Return, and Late Return. These flowcharts are used to show the step-by-step procedures and decision-making processes within the system.

Figure 1 below shows the main workflow of Hasta Car Rental System, starting from process user registration and login until the booking completion. It illustrates the main path for user roles including customer and staff and shows how different processes connect within the system.

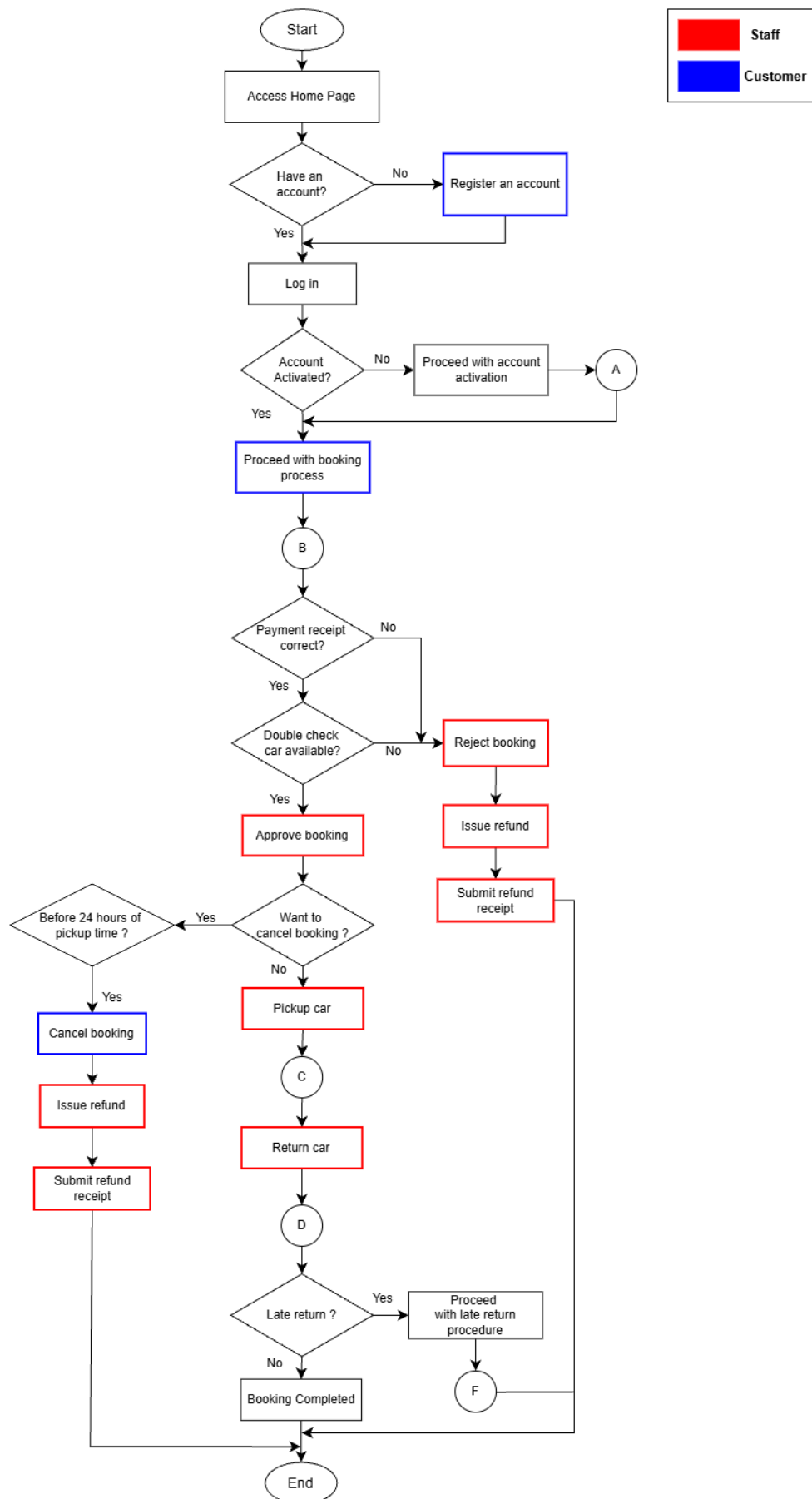


Figure 1: Overall Flowchart

Figure 2 above demonstrates the details of the account activation process. Customers need to provide all the personal information and submit required documents for staff to review them. Staff will approve or reject the activation after verifying the documents. Upon approval, customers can get full access to make booking. If the activation is rejected, customers can review the disapproval reason and submit a new activation request.

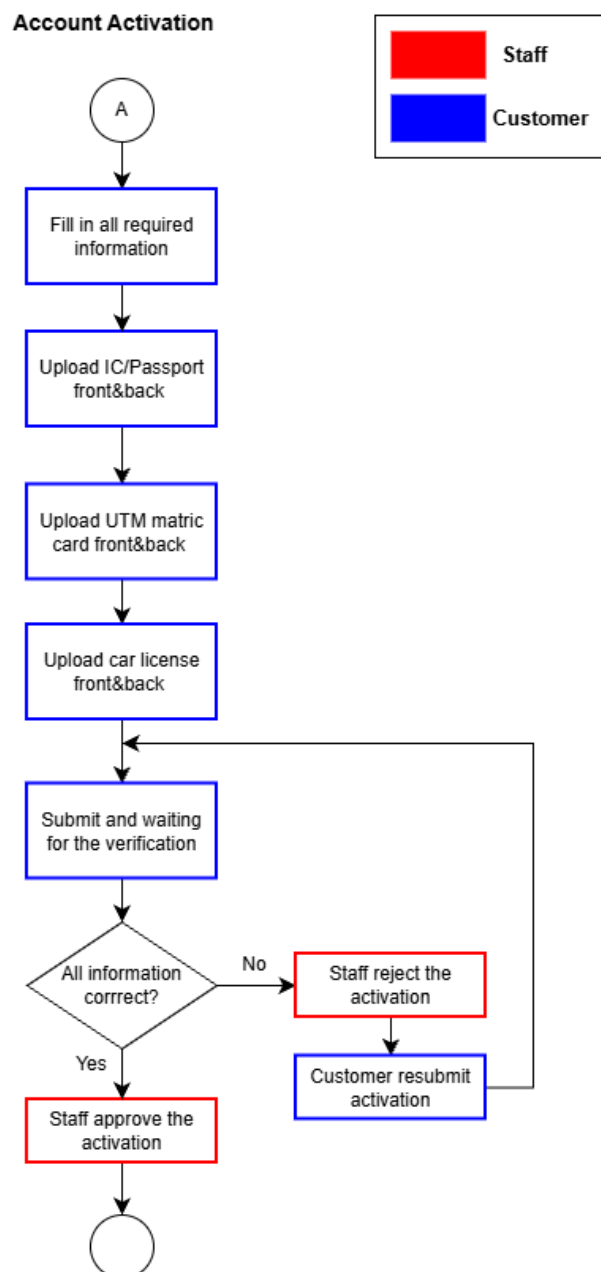


Figure 2: Account Activation Flowchart

Figure 3 shows the booking process flows starting from select pickup and return details. After that, customers can select available cars that are filtered based on the booking details. A delivery fee will be required if the pickup or return or both are outside of UTM. Then customers can choose to use a voucher if available and complete the payment. The submitted booking needs to wait for staff to verify the payment and approve the booking.

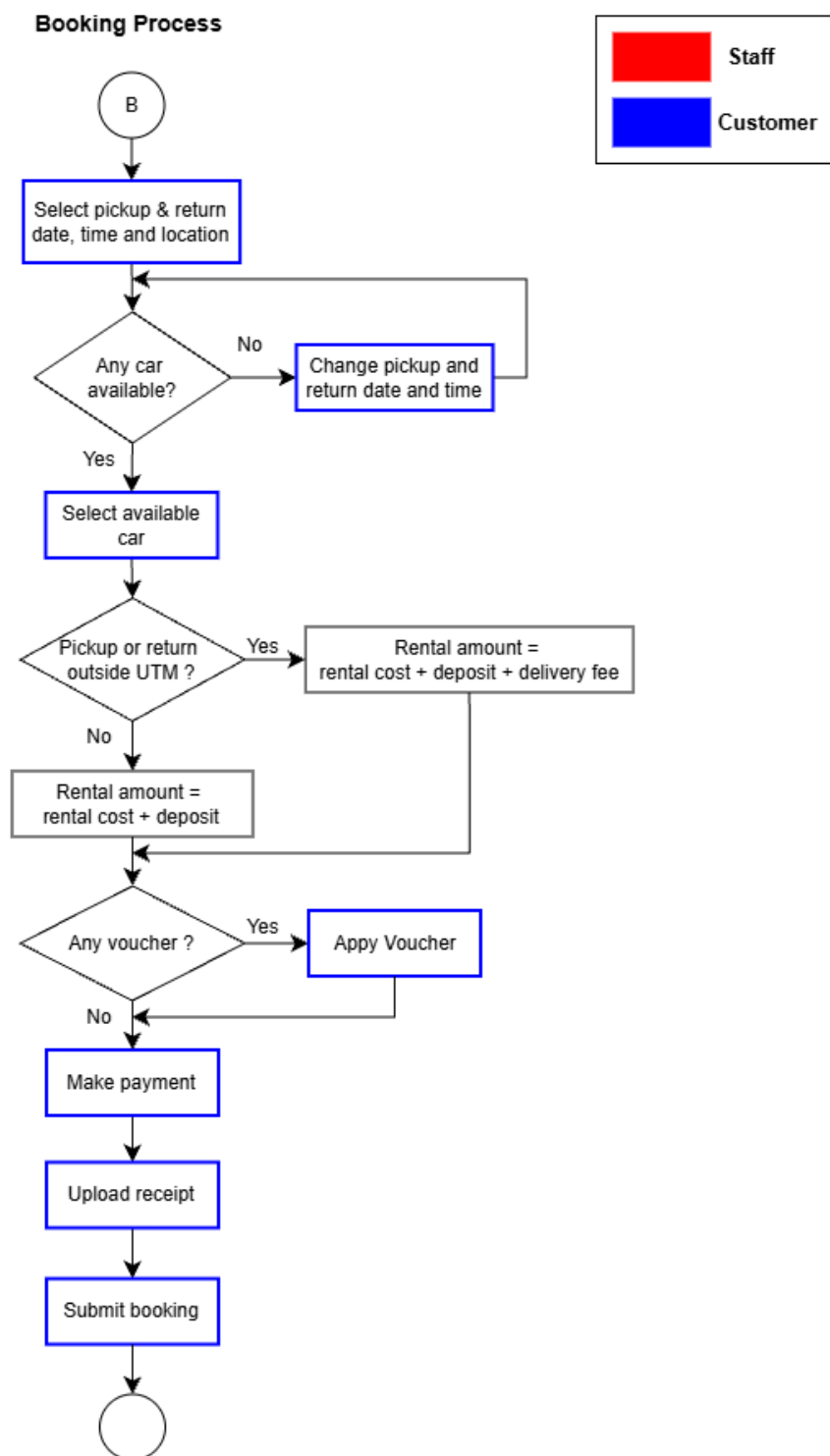


Figure 3: Booking Process Flowchart

Figure 4 shows the car pickup process where customers can only pickup 30 minutes before the scheduled pickup time. Customers must download the agreement and inspection form, sign them, and then upload back to the system. Next, they need to upload five car photos including front, back, left, right view and the car odometer before submitting the form.

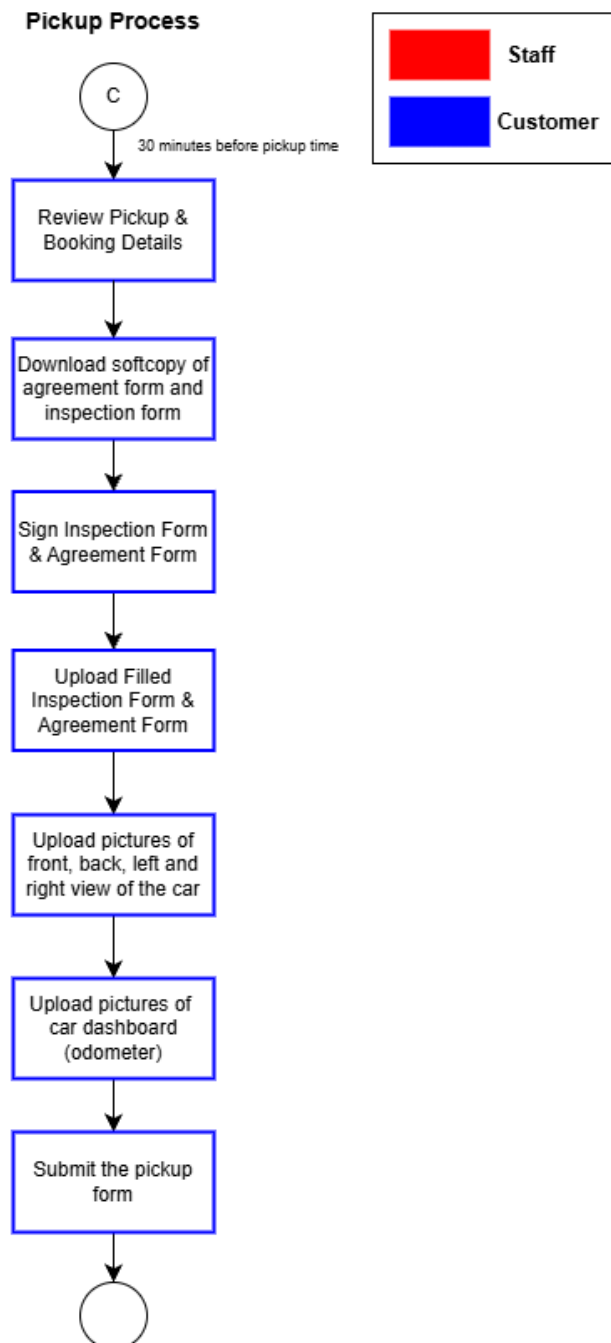


Figure 4: Pickup Process Flowchart

Figure 5 outlines the on-time car return process where customers need to upload six photos (including photo of car key), report an issue if faced any problem during rental, provide feedback and provide bank details in order to get the deposit refund. After the car return, there are some tasks needed to be completed by the staff including checking customer feedback, resolve issue if any, update car mileage and fuel bar, as well as issue deposit refunds.

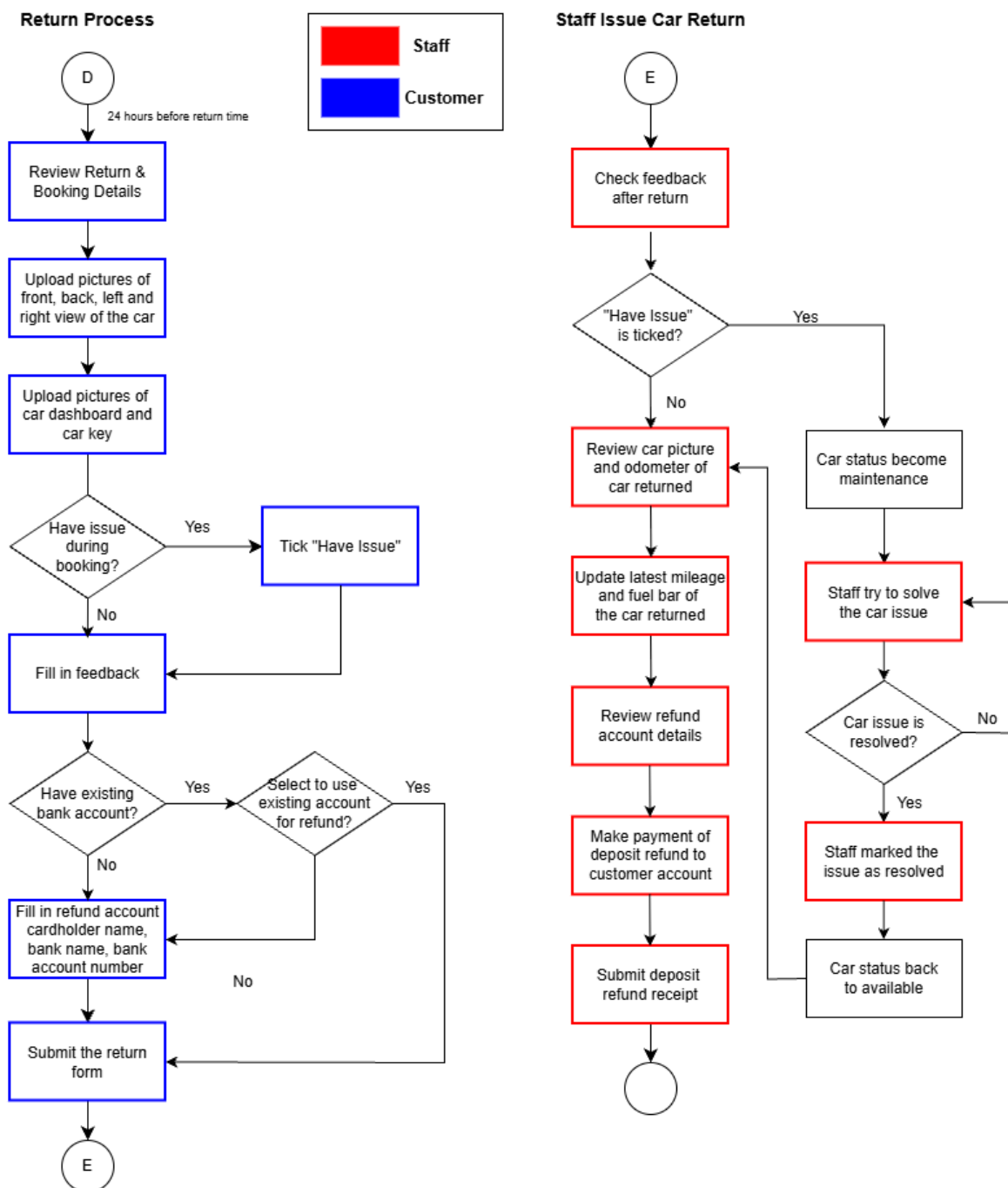


Figure 5: Customer Return Car and Staff Issue Return Flowchart

Figure 6 shows the late return process if the car is returned more than 30 minutes after the scheduled return time. Customers will be charged a penalty of RM50 per hour and their accounts will be automatically blacklisted. This causes them to not do any booking as their accounts are suspended until the penalty payment is approved by the staff.

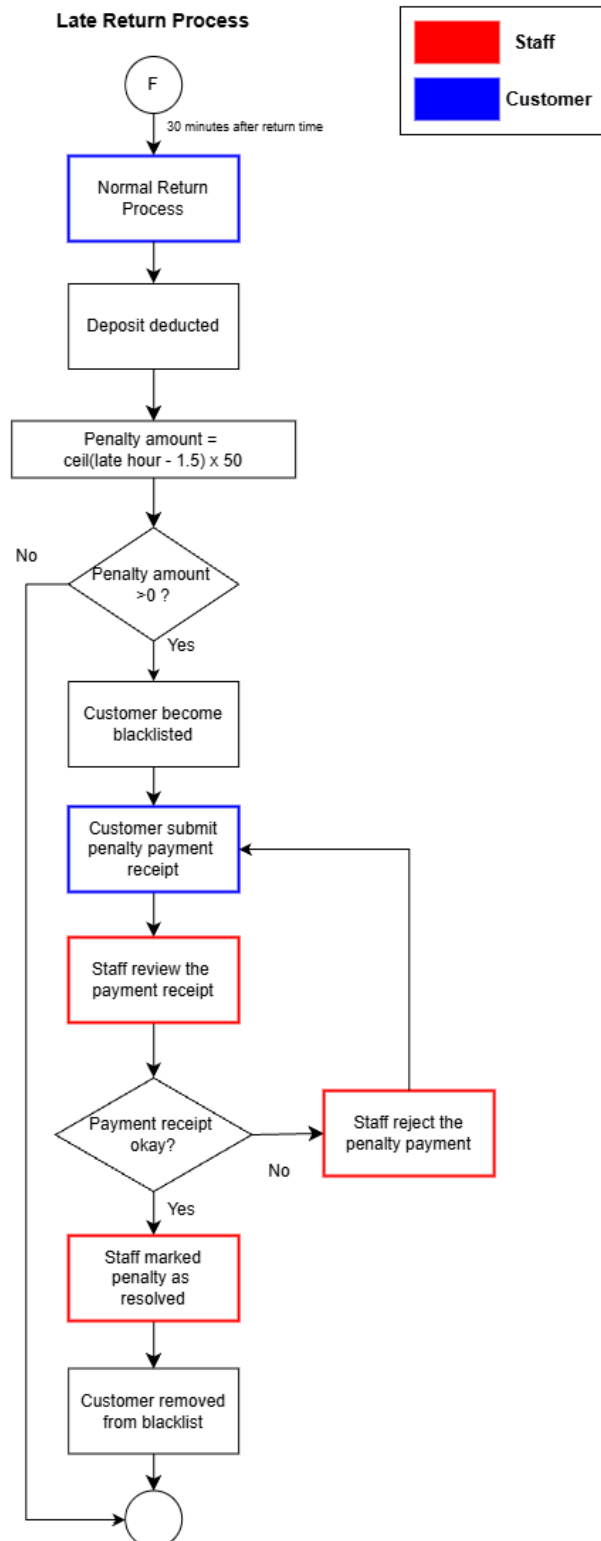


Figure 6: Late Return Flowchart

1.4.2 Proposed System Sitemap

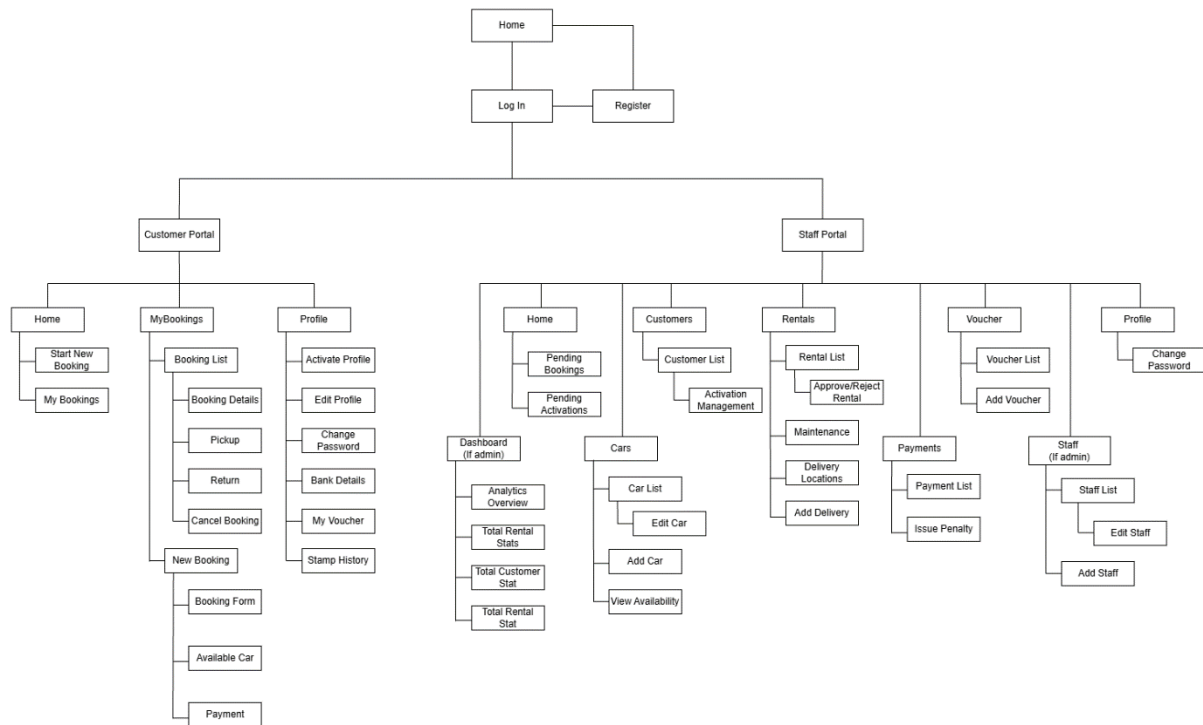


Figure 7: Proposed System Sitemap

Figure 7 above is the sitemap that shows the hierarchical structure and navigation flow of the Hasta Car Rental System. It shows all the accessible pages organized by user roles which are Customer, Staff and Admin.

1.4.3 Use Case Diagram

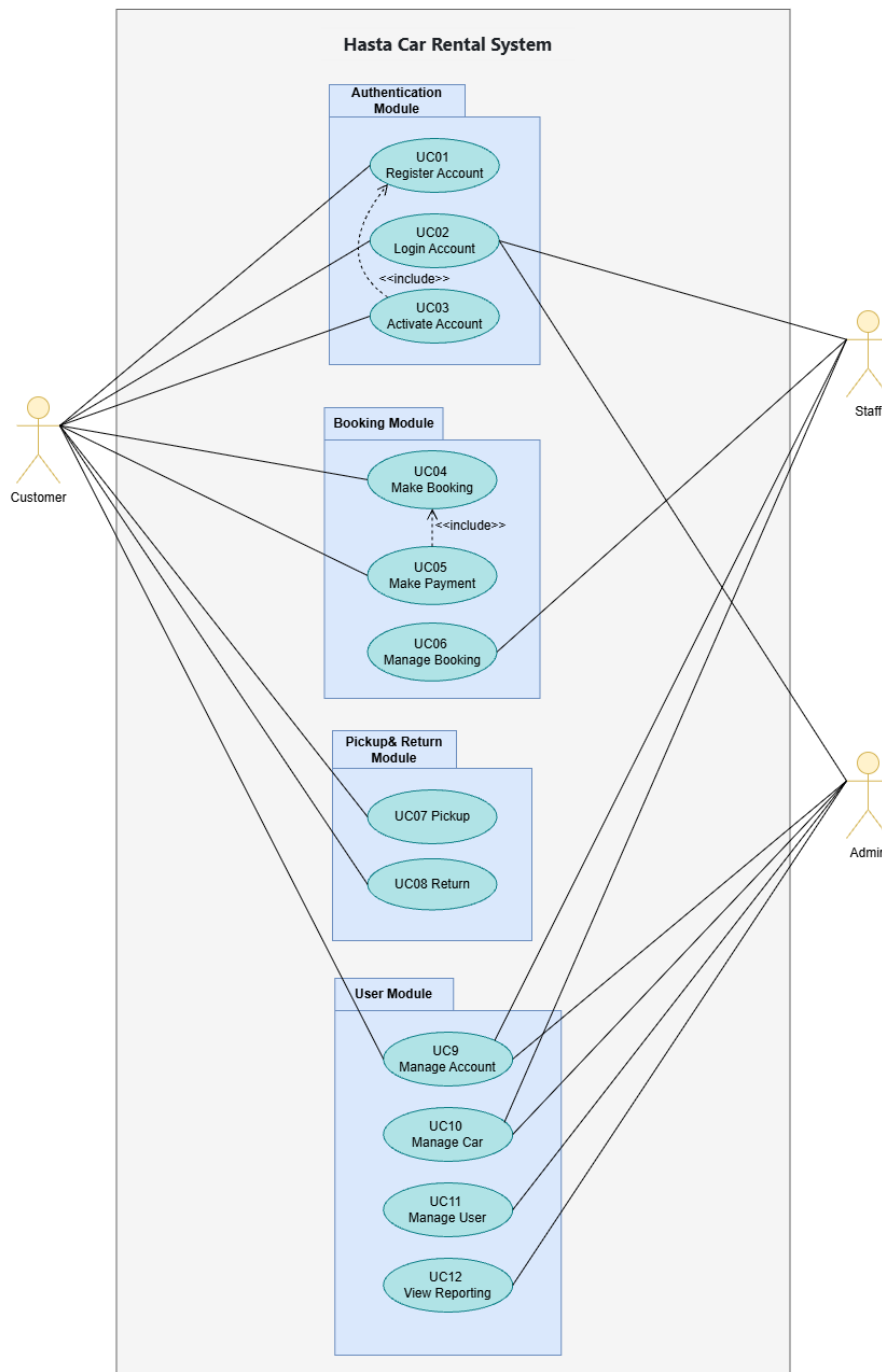


Figure 8: Use Case Diagram for Hasta Car Rental System

Figure 8 above shows the use case diagram for the Hasta Car Rental System with three actors: Customer, Staff, and Admin. Customers can register, login, activate accounts, make bookings, submit payments, pickup and return vehicles, and manage their accounts. Staff handle account activation, booking management and car management. Admins manage users, cars, and view system reports.

Use Case	Function	Description
UC01	Register Account	This use case allows customers to sign up as a user for the system.
UC02	Login Account	This use case allows users to log in to the system using their email and password.
UC03	Activate Account	This use case allows customers to activate accounts by submitting required documents then staff reviews and approve activations.
UC04	Make Booking	This use case allows customers to create a booking by specifying pickup and return details and selecting available cars.
UC05	Make Payment	This use case allows customers to complete payment by uploading the payment receipt and submitting booking.
UC06	Manage Booking	This use case allows staff and admin to view all bookings, approve or reject the bookings.
UC07	Pickup	This use case allows customers to process car pickup by completing the agreement and inspection form and capturing pickup car photos.
UC08	Return	This use case allows customers to return the car by capturing return car photos including car key photo, providing feedback and reporting an issue that occurred during the rental period (if any).
UC09	Manage Account	This use case allows all users to view their own profile information and change account password.
UC10	Manage Car	This use case allows staff and admin to add new vehicles to the rental fleet and update existing car information.
UC11	Manage User	This use case allows the admin to view all user accounts in the system, create new accounts for staff and update user status.
UC12	View Reporting	This use case allows admin to view comprehensive reports including rental statistics, revenue reports and customer analysis.

1.4.4 Feasibility Study

I) Technical Feasibility

Current Technical Situation:

- Booking records are stored across WhatsApp messages, Excel spreadsheets, manual whiteboards, and Wahdah system.
- No centralized database exists for real-time data synchronization.
- High risk of data redundancy, data inconsistency, and delayed updates.

Proposed System Feasibility:

- The system can be developed using a relational database management system (RDBMS) such as MySQL or PostgreSQL.
- The centralized database enables real-time booking status updates, automated availability checking, and structured data storage for bookings, vehicles, payments, refunds, and loyalty records.
- It integrates with payment verification records and document upload and storage.
- Required hardware (PCs, internet access) already exists in the organization.
- Staff already use digital systems (Excel, Wahdah), reducing the learning curve.

The proposed system is technically feasible, as it relies on widely available, stable, and well-supported database and web technologies that match the company operational scale.

II) Operational Feasibility

Current Operational Situation:

- Staff workload is high due to repeated manual data entry.
- Same data is entered multiple times (WhatsApp, Wahdah, Excel).
- Human errors occur in checking booking availability and payment verification.
- No real-time visibility for management reporting.
- Refund processing is slow and manually tracked.

Proposed System Feasibility:

- A centralized database eliminates duplicate data entry.
- Automated validation of booking eligibility, pickup and return time constraints.
- Role-based access ensures staff manage bookings and management views reports.
- Automated workflows for return processing, mileage updates, penalty calculation, and refund status tracking
- The reporting module supports daily sales, vehicle utilization, and loyalty analysis.

The system is operationally feasible, as it simplifies staff tasks, improves data accuracy, and aligns well with existing workflows while reducing manual effort.

III) Economic Feasibility**One-Time Development Cost:**

- System Design & Development: RM 18,000 - RM 25,000
- Database Design & Implementation: RM 5,000 - RM 8,000
- Testing & Deployment: RM 3,000 - RM 5,000
- Staff Training: RM 1,000 - RM 2,000
- Total One-Time Cost : RM 27,000 - RM 40,000

Recurring Operational Cost (Per Year)

- Server Hosting & Database: RM 1,500 - RM 3,000
- Maintenance & Updates: RM 3,000 - RM 5,000
- Backup & Security: RM 1,000 - RM 2,000
- IT Support (Part-time): RM 4,000 - RM 6,000
- Total Annual Cost: RM 9,500 - RM 16,000

Although the proposed system requires an initial investment of approximately RM 27,000 to RM 40,000, the reduction in staff workload and operational inefficiencies can offset the cost within 1 to 2 years. The recurring annual cost is relatively low compared to the long-term operational savings and improved data reliability.

1.5 Objectives

1. Digitalize and Automate Workflow for Each Booking.
2. Reduce Manual Workload and Repetitive Data Entry.
3. Centralize Data Management in Organized and Systematic Database.
4. Enhance Booking Accuracy and Customer Experience.
5. Improve Task Efficiency and Operational Control of Staff.
6. Support Future Scalability for Business Growth.

1.6 Project Scope

The scope of this project is the database analysis and design phase of the car rental system. This includes the study of current transaction processes, identifying the data requirements, defining entities, attributes and relationships, and specifying the transaction type like data entry, data updates, data delete, and data queries. This project should also cover the clear understanding of the business rules and the system boundaries, such as the major user views, internal system, and external system. Besides, a proposal draft is also given to be reviewed and used in the design phase if approved.

1.7 System Boundary

The system boundaries define the scope of the Hasta Car Rental System by differentiate the internal functions that are automated and handled within the system, and also the external operations that must be performed manually or outside the system. This explanation would provide a clear picture of capabilities of the system, limitations of the system and the responsibilities of the users.

1.7.1 Internal Scope

i) General access for all users

Page	Function
Register	• Customer account registration with form validation • Automatic email uniqueness check • Automatic NEWUSER10 voucher for first-time customers
Login	• User authentication for all roles including Customer, Staff and Admin • Role-based redirection to appropriate portal

ii) Customer Portal

Page	Function
Home	• Overview of account status and activation progress • Quick access to create booking function • Display of important notifications and reminders
Booking List	• View booking history and current bookings • Access to booking details including car and booking information, payment summary and booking timeline • Cancel booking

Make Booking	• Create new booking • Apply Voucher • Upload payment receipts • Receive email notifications once booking approved
Pickup	• Download agreement and inspection form • Upload signed forms • Upload 5 car photos (front, back, left, right, odometer) • Can upload additional 1 photo if found damage on car • Receive email reminder for pickup time
Return	• Upload 6 car photos (front, back, left, right, odometer, car key) • Report any issue encountered during rental • Submit feedback on rental experience • Provide bank details for deposit refund *For late return: • View penalty amount • Submit penalty payment with the receipt • Wait for staff approval
Profile	• View and edit personal information • Submit activation request if new user • Save bank detail for return use • View available vouchers • View stamps history • Change account password

iii) Staff Portal

Page	Function
Dashboard (Admin only)	• View rentals, customers, revenue analytics with time period filters • View booking trends graph, income growth chart, customer growth chart, customer distribution by faculty and college, and booking heatmap
Home	• Overview of pending tasks • Quick access to all management functions
Car	• View complete car list with details • Add new car • Edit car details • Change car status via dropdown menu • View all rental history for each car • Check car availability for specific dates
Customers	• View all customer list • View pending activation requests • Review request and choose to approve or reject the activation • Change customer status via dropdown
Rentals	• View all rental records • Access pending rentals for approval • Review all booking details and verify receipt • Approve or reject bookings • View car flagged for maintenance • Check customer feedback to manage issue • Change car status to Available after resolved issue • View pickup and return car photos • Issue deposit refund • Export rental reports to PDF or Excel format

Delivery Locations	• View list of available pickup and return locations • Edit details of delivery locations • Add new delivery locations
Payments	• View list of all payment record with receipt • Verify penalty payments and remove customers from blacklist • Export payment reports to PDF or Excel format
Vouchers	• View all vouchers usage status and usage history • Issue voucher to customers
Staff (Admin only)	• View list of all staff with details • Edit staff details • Add new staff to create account for them
Profile	• Change password

1.7.2 External Scope

Process	Manual Operations
Authentication and Verification	• Manual verification of the authenticity of IC/Passport, UTM cards and driving license that uploaded by customer • Cross-check document information with the registered data. • License expiry validation
Payment	• Manual verification of payment receipts with the bank statements • Manual refund processing to customer
Pickup	• Customers download the forms and sign them before uploading to the system. • Manual verification of the uploaded signed agreement and inspection form • Staff manually send car to the pickup location if not at Hasta
Return	• Physical car inspection by staff • Physical car collection if not at Hasta
Maintenance	• Scheduled car maintenance or repairs by third-party

1.7.3 Major User Views

i) Customer View

The customer portal gives functions to customers to manage their account and create a booking. They can create an account and log in, activate, and update their account. They can create a new booking by filling in booking information and choosing a car and pay. After that, they can see booking history and booking details, cancel their bookings at least 24 hours before pickup time, and track the stamps and vouchers received in their profile.

ii) Staff View

The staff portal supports various operational activities necessary to manage the car rental services. Staff can log in to the system to view pending bookings, pending customer activations, manage customer account and activation status. In addition, staff can also manage cars through add or edit cars, track returns of cars, rental history of cars, approval, or rejection of rentals, update the maintenance and resolve issues reported. Moreover, staff can also manage delivery locations and vouchers, issue refunds and issue penalties.

iii) Admin View

Admin view is an extended staff view that comprises system-level monitoring and administrative control. Admin can access various dashboard analytics such as total rental and customer statistics as well as managing staff accounts by viewing the list of staff and performing add/edit staff operations.

1.8 Project Planning

1.8.1 Human Resources

Team Member	Software Engineering	Database	System Development Technology
Chun Yao Ting	Project Manager - Plan and manage the project schedule. - Coordinate tasks among team members and monitor progress. - Ensure project objectives and requirements are met.	Database Administrator - Manage and maintain the database system. - Ensure data security, backup, and recovery. - Monitor database performance and reliability.	Head Programmer - Lead the system development and coding activities. - Assign programming tasks to developers. - Ensure code quality and system integration.
Evelyn Ang	Developer - Develop system features based on requirements. - Implement and integrate system functionalities. - Fix bugs and improve system performance.	Database Designer - Design database schema and table structures. - Define relationships between database entities. - Ensure database normalization and efficiency.	Developer - Write and implement system source code. - Integrate system components and features. - Support debugging and code improvement.

Mirza As-Siddiq Bin Tohari	Quality Assurance - Tests system functionalities to ensure correctness. - Identify and report defects or issues. - Verify that the system meets requirements and quality standards.	Data Analyst - Analyze data using SQL queries. - Generate reports and data summaries. - Validate data accuracy and consistency.	Tester - Perform functional and system testing. - Detect and report system errors or bugs. - Assist developers in debugging and testing fixes.
Muhammad Aliff Bin Helmi	Documentation - Prepare project documentation and reports. - Maintain system descriptions, diagrams, and manuals. - Ensure documentation is clear and updated.	Data Validator - Check data accuracy and completeness. - Validate data consistency and constraints. - Ensure correct data input and storage.	Developer - Implement system features and functionalities. - Assist in system debugging and testing. - Support code integration and refinement.

1.8.2 Work Breakdown Structure

1. Project Initiation & Planning

1. Define Project scope and objectives
2. Identify stakeholders

2. Current System Analysis

1. Documenting existing car rental workflow
2. Identify manual data handling methods
3. Identify pain points

3. Proposed System Requirements

1. Define user roles: Customer, Staff, Admin
2. List functional requirements per module: Customer, Booking, Admin
3. Define non-functional requirements

4. Database and UI Planning

1. Design ER diagram and schema
2. Create wireframes

5. Development & Delivery

1. Customer Module

1. Customer Registration and Login
2. Profile Management and Document Upload
3. Document Verification by Admin

2. Booking Module

1. Car Listing and Availability
2. Booking Form and Status Management
3. Payment and Receipt Upload
4. Payment Validation by Admin
5. Loyalty System
6. Pickup and Return Process
7. Debt and Blacklist Management

3. Admin Module

1. Staff CRUD
2. Booking Management
3. Financial Panel
4. Vehicle Maintenance and Status

6. Integration and Testing

1. End-end System Integration
2. UAT and Bug Fixing
3. Deploy Demo and Final Checks

7. Final Documentation and Presentation

1.8.3 Gantt Chart

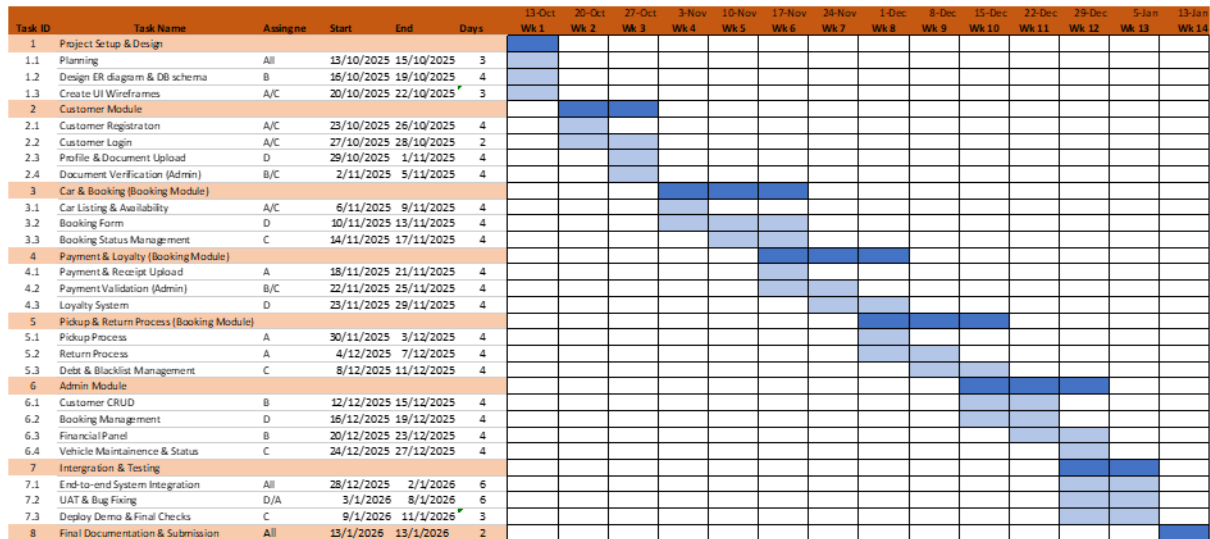


Figure 9: WBS Gantt Chart of Project Planning

1.9 Requirement Analysis

a) Start Booking

The current operation mode for booking in Hasta Travel & Tours Sdn. Bhd. is mostly manual, meaning that staff have a heavy workload. To book a car, customers had to contact Hasta Travel & Tours Sdn. Bhd. by WhatsApp message. Then, the staff will generate an immediate response to prompt the user to fill in the information such as are you a UTM student or staff, date, time, and location for pickup and return purpose, and finally the destination. It also notified the customers about the rule of minimum booking of 2 days for destinations outside Johor. This first message is for staff to check the availability of the car. Next, after the customer fills in the information, staff have to confirm the customer is a UTM student or staff, check whether the pickup location and return location is in UTM Student Mall, and check whether the pickup and return time are within office hours. (9am-5pm) If the customer is not a UTM student or staff, the booking cannot be continued. If the location is outside UTM or the time is not within working hours, it requires the staff on duty to determine how much charge will be given for the customer, such as delivery fee.

Then, after checking the availability of the car through the availability whiteboard or through Wahdah, the staff will inform the customer about the available cars and their respective price. Next, after confirming the car selected, staff will prompt the customer again to fill in the personal details such as IC or passport number, phone number, matric number, race, college, faculty, email. The message also gives deposit refund details and requires the customer to give a payment receipt for the deposit of RM50. If the pickup or return location is outside UTM, and pickup or return time is out of working hours, the PIC will be given as well, and staff will notify the customer of the extra charge as well. Next, staff will verify the total amount and whether the payment date and time in the receipt is aligned with the bank statement. After deposit verification, customers need to send the front and back picture of IC/Passport, driving license, and matric card with white background. After verification of documents, the customer side procedure is completed, and the staff will mark the date, pickup time, return time and customer name in the availability whiteboard. Next, the staff will manually key in all the information at Wahdah staff system, then finally the rental information is recorded. To record the new booking, they also store the rental information in daily sales reports in Excel format. This means that staff also need to manually key in all the required information.

b) Car Pickup

To handover the car key during pickup, first the staff will update the booking information if it is needed. (e.g. change the status from “Unpaid” to “Paid”) Then, they need to download an inspection form and agreement form, which will be automatically filled with all the information recorded in the Wahdah system. If the pickup is via online, the document will be sent to the customer for checking and signature before 30 minutes of booking time, with the pickup and return procedure message attached. Customers will sign the form and send it back to Hasta Travel through WhatsApp, with the picture of the cars and payment receipt attached. Afterwards, staff have to print the inspection form, agreement form, and verification documents to store as hardcopy. They also need to manually write down the mileage, fuel level, date, time and PIC, as this information is not automatically filled in the inspection form generated. If the pickup is physical, staff just need to print out the form and ask the customer to fill in on spot and have the inspection session physically. The sales report is also updated.

c) Return Car

To track the return status of the car, Hasta Travel uses Tracksolid Pro to track the location of the car, as they had their GPS installed on the rented car. Staff will select the car they wish to track, with the return date and time. Then, the location of the car will be shown. After the customer returns the car, staff will record the mileage and fuel level after booking. If the return is late, the return penalty will be counted starting from 30 minutes of return time, and staff will put remarks on the agreement hardcopy form. If there is damage to the car, staff will enquiry customers on the reason of damage, with the comparison of the car picture before booking. Some penalties could be imposed on customers. If there is no issue, staff will send the car picture after returning to the company, and the deposit will be refunded in 10 days. After returning the car, the information is also updated in the Excel sheet of the sales report.

d) Loyalty Card Feature

The loyalty record will be recorded by staff in the Excel sheet. To get the first stamp, the customer had to book at least 12 hours for any type of car. To get the stamp afterwards, the customer had to book at least 9 hours for any type of car. After giving the loyalty card to a customer, record the customer’s name according to the card’s serial number. Mark tick in the Excel sheet and write down the date and the name of the person in charge. If the customer rents again for the next 9 hours and has reached the box that provides a discount, mark the “Claimed” button to indicate that the discount has been redeemed.

1.9.1 Current Business Workflow Flowchart

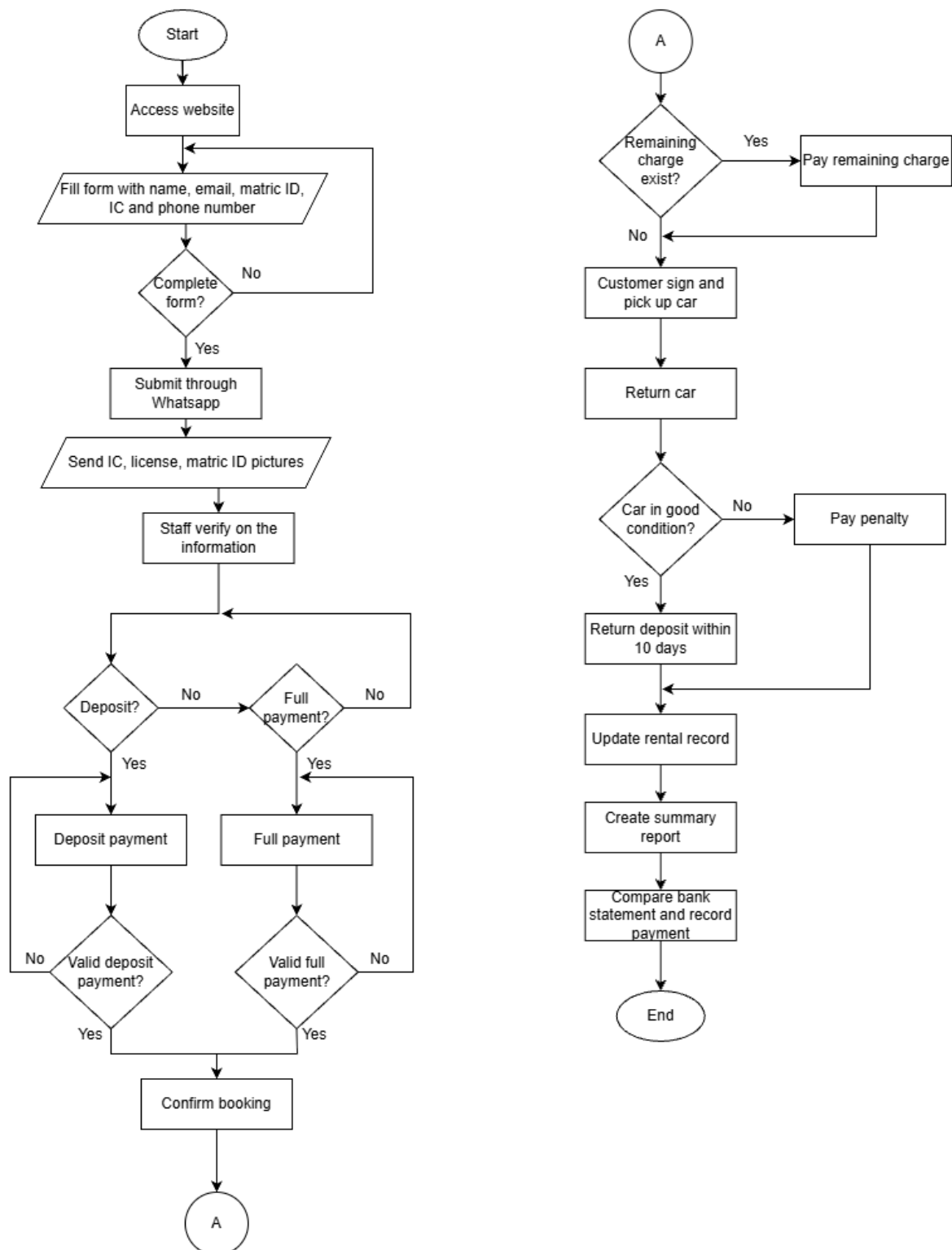


Figure 10: Current Business Workflow Flowchart

1.9.2 Comparison with Existing System

	Existing System	drive.my	GoCar
Verification Form	Customers are required to send a copy of their IC/Passport, driving license, and student/staff card 3 days before their booking session. The verification is done by the admin through WhatsApp manually.	There is no verification done on this website.	After signing up an account, the system requires customers to verify their identity by uploading their IC, driving license and a selfie of the customer.
Registration form	In the existing system, personal information is given through WhatsApp, and the admin will book the session. There will be no account created for the customer at this point.	A simple registration consists of username, full name, email address, password, and their agreement with drive.my terms and privacy policy.	A simple registration consists of full name, email address, password, and their agreement. It also has a direct sign-up option for google accounts.
Booking form	All bookings will be confirmed by the admin manually through WhatsApp. For pickup and return done within office hours (9am - 5pm), and pickup and return point at UTM Student Mall, there will be no additional charges. However, for bookings done outside office hours, or pickup and return point other than UTM Student Mall, a staff member will be needed to handle the customer. The booking cannot proceed until there is a staff member available to handle the customer. The admin will confirm with the staff on duty the amount to be charged and inform the customer about the applicable charges.	After filling up pickup, return location, and the date, the list of cars available can be chosen by the customer. After choosing the designated vehicle, there's a list of add-ons that can be made by the customer. The info of pickup and return will be shown as well as the total of payment of the booking alongside it. After confirming the price, the customer can proceed with the payment.	The customer needs to choose their booking dates and time. After filtering the dates, the available car with a price can be chosen by the customer. The chosen car will be shown where we need to pick it up, how long the duration of the booking along with its price.

Payment form	Customers need to inform the booking price of the car to the staff incharge. The customer needs to pay an additional RM 50 as deposit to the staff. The online payment by the customer needs to come with a receipt from the bank as proof of payment and it will be put on Wahdah by the staff manually. If the customer pays with cash, the staff needs to bank the cash inside his/her own account and immediately transfer it to the HASTA account. This transaction needs to be with a receipt as it will be attached inside DSR.	The customer needs to enter their billing information such as their full name, email address, address, and preferred payment method (Direct bank transfer or Payex). If the customer chooses direct bank transfer, the system will send an email, and the booking will be on-hold until the payment has been received. If the customer chooses Payex, they will have four different options to choose from (card, online banking, eWallet, and instalment).	After selecting the car, customers are required to choose one of Collision Damage Waiver(CDW) options and one choice of flex package. The CDW will affect the amount that needs to be paid when a collision occurs during your ride. The flex package includes the basic package and Premium Flex. Premium Flex package will provide the customer flexibility to return the car 30 minutes late, refund for cancellation via GoPocket points, and unlimited change of booking date and time. The total payment consists of subtotal, CDW, flex, and tax. There will be no deposit needed. The customer can choose their preferred payment method (GoPocket, credit/debit card, online banking, and eWallet).
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Deposit payment	After filling up the booking form, the customer needs to pay forward the deposit payment. The payment must include a receipt that has the same date as the day the payment was made, and the deposit amount must be correct. Deposit payment is a must as it ensures the booking can be made. The deposit will be returned in 10 working days if: • No damage to the car & the car is in good condition, same as it was. • Off returned same as given. • No summons during your rental • Car is in clean condition. • Return on time.	Customers are required to pay a refundable deposit depending on which car model. The deposit prices range from RM300 to RM 1000.	No deposit needed to be paid.
Filter features	No filter features.	These filters help customers to find suitable price ranges, vehicle categories and vehicle brands that suit the customer preference.	This app allows customers to filter the car type, number of seats, and model that they want. This will improve the usability of the application for the customer experience.

1.9.3 Database Requirements

a) Performance Requirements

- The database must support frequent read and write operations, especially during booking, pickup, and return periods.
- Booking availability checks must be handled quickly to avoid double-booking of cars.
- Retrieval of customer records, vehicle records, and past bookings must be efficient to support daily operations and reporting.
- The system must support historical data storage for tracing past rentals, payments, and loyalty usage.
- Query should take less than 1 seconds, even under high traffic scenario.

b) Data Consistency & Integrity Requirements

- Customer, vehicle, booking, and payment data must remain consistent across all records.
- Each booking must be linked to one valid customer and one valid car.
- Payment records must be associated with the correct booking.
- Duplicate customer and booking records should be avoided.
- Booking time periods must not overlap for the same car.

c) Security Requirements

- Sensitive data such as IC/passport numbers, license numbers, and bank details must be securely stored.
- Only authorized staff should be allowed to view customer documents, approve bookings, process refunds.
- Different access levels are required, where customers will view their own data, staff will manage bookings and admin will manage staff and reports.

d) Availability & Reliability Requirements

- The database must be available during office hours (9am–5pm) at minimum.
- Data must not be lost during booking, payment verification, or document submission.
- Backup and recovery mechanisms are required to protect against accidental data loss.

e) Scalability Requirements

- The database must support future growth in the number of customers, vehicles, historical booking records and transaction records.
- Expected number of customers are 1000, with expected growth of 10% each year.
- Expected number of cars are 20, with expected growth of 10% each year.
- Expected number of bookings are 2000 per month, 10000 for historical bookings, with expected growth of 10% each year.
- Expected number of transactions are 2000 per month, 10000 for historical transactions, with expected growth of 10% each year.
- Database design must allow future expansion without restructuring existing data.

1.9.4 Data Requirements

a) Customers:

Attributes:

1. Customer ID
2. Full name
3. IC or Passport Number
4. UTM Role (Student/Staff)
5. Matric Number (Student or Staff ID)
6. Race
7. College
8. Faculty
9. Phone Number
10. Email
11. Status (Active / Blacklisted)

Relationship:

One customer can make many bookings.

b) Vehicles:

Attributes:

1. Car ID
2. Car Plate
3. Car Model
4. Price Rate
5. Status (Available, Booked, Maintenance)
6. Mileage

Relationship:

One car can be used for many bookings.

c) Staff:

Attributes:

1. Staff ID
2. Full Name
3. Role
4. Email

d) Bookings:

Attributes:

1. Booking ID
2. Customer ID
3. Car ID
4. Pickup Date
5. Pickup Time
6. Return Date
7. Return Time
8. Pickup Location
9. Return Location
10. Status (Pending, Approved, Picked Up, Returned)
11. Total Amount

Relationship:

Booking must belong to one customer.

Booking must reserve one car only.

Constraint:

Booking time must not overlap with another booking time for the same car.

e) Payments

Attributes:

1. Payment ID
2. Booking ID
3. Payment Type (Deposit, Penalty, Instalment, Full Payment)
4. Amount
5. Payment Date & Time
6. Receipt
7. Verification Status

Relationship:

One booking can have many payment records. (Deposit, Penalty, Instalment, Full Payment)

f) Inspections

Attributes:

1. Inspection ID
2. Booking ID
3. Mileage
4. Fuel Level
5. Agreement Form
6. Inspection Form
7. Pickup Date & Time
8. Return Date & Time

1.10 Transaction Requirement

1.10.1 Current Transaction Requirement

1. Customer Booking Request

Data Entry:

- Staff manually records customer booking details from WhatsApp, which are customer name, UTM status (Student / Staff), pickup & return date and time, pickup & return location, and destination.

Data Update / Delete:

- Staff manually edits chat messages or information if a customer changes details.
- No structured delete so old booking requests remain in chat history.

Data Query:

- Staff manually reads WhatsApp chat history to retrieve booking details.
- Staff checks customer must be UTM student or staff, and minimum booking period for destination outside Johor.

2. Car Availability Checking

Data Entry:

- Staff manually marks booking dates on the availability whiteboard.

Data Update / Delete:

- Staff erases or overwrites whiteboard records when availability changes.

Data Query:

- The staff visually checks the whiteboard.
- Staff logs into the Wahdah system to cross-check car availability.

3. Customer Personal Information Collection

Data Entry:

- Staff requests personal details via WhatsApp, which are IC/passport number, phone number, matric number, race, college, faculty and email address.
- Information is manually typed by staff.

Data Update / Delete:

- Personal information updates or corrections require clarification via WhatsApp.

Data Query:

- Staff scrolls through chat history to retrieve customer information.
- Previous customer data cannot be reused.

4. Payment & Deposit Submission

Data Entry:

- Customer sends payment screenshot via WhatsApp.
- Staff manually records deposit payment (RM50) details.
- Receipt information is stored in chat or printed if necessary.

Data Update / Delete:

- Staff manually notes payment confirmation.
- No structured delete as old receipts are kept manually.

Data Query:

- Staff manually compares receipt date and time with bank statements.
- Staff searches chat history for receipt.

5. Document Verification

Data Entry:

- Customer sends pictures of IC/passport (front & back), driving license and matric card.
- Staff saves files locally or prints them.

Data Update / Delete:

- Documents are re-requested if unclear.

Data Query:

- Staff manually checks files or printed documents.
- Staff searches folders or WhatsApp messages.

6. Booking Confirmation & Recording

Data Entry:

- Staff writes booking details on availability whiteboard, Wahdah system, and Excel daily sales report.

Data Update / Delete:

- Staff manually edits Wahdah and Excel if booking changes.
- Whiteboard records are erased or overwritten.

Data Query:

- Staff checks Excel for daily sales.
- Staff checks Wahdah for booking details.

7. Car Pickup Process

Data Entry:

- Staff downloads and prints inspection form and agreement form.
- Mileage, fuel level, date, time, and PIC are handwritten.
- Pickup photos and signed forms are collected.

Data Update / Delete:

- Booking status updated manually in Wahdah from unpaid to paid.

Data Query:

- Staff retrieves booking info from Wahdah.
- Staff refers to printed documents for inspection history.

8. Car Return Process

Data Entry:

- Staff records return mileage and fuel level manually.
- Staff takes return photos and prints them.
- Remarks written on agreement form if late or damaged.

Data Update / Delete:

- Staff updates Excel sales report.
- Penalties recorded manually on hardcopy forms.

Data Query:

- Staff compares pickup and return photos manually.
- Staff checks GPS location of car via Tracksolid Pro.

9. Penalty & Deposit Refund Handling

Data Entry:

- Staff records late return penalty manually.
- Refund details are recorded manually via Excel.

Data Update / Delete:

- Manual change of penalty amount or refund amount.

Data Query:

- Staff checks bank statements manually.
- Staff checks Excel for refund status.

1.10.2 Proposed Transaction Requirements

1. Authentication Process

Data Entry:

- Register new account by customer.
- Create staff account by admin.
- Submit activation request with uploaded documents.
- Auto-issue NEWUSER10 voucher upon registration

Data Update/Delete:

- Update user password.
- Update customer status to “Approved” or “Reject” after review activation request.
- Update customer status via dropdown (New, Pending, Active, Inactive, Blacklisted)
- Update staff information (Full name, email, position)
- Delete staff account.

Data Query:

- View pending activation requests.
- Retrieve customer details and uploaded documents.
- Retrieve user role for dashboard redirection.

2. Booking Process

Data Entry:

- Create new booking with selected dates, time, locations and car.
- Submit payment receipt.
- Apply voucher to booking.
- Set initial booking status to “Pending”.

Data Update/Delete:

- Update booking status to “Approved” or “Rejected” after review booking detail and payment receipt.
- Update car status to “Rented” if approved.
- Upload refund receipt for rejected and cancelled booking.
- Cancelled booking and update booking status to “Cancelled”.
- Update voucher usage to “Used”.

Data Query:

- Check car availability for selected dates and time.
- Retrieve car details.
- View all pending bookings.
- Retrieve booking details for review.
- View all payment lists.

3. Car Pickup Process**Data Entry:**

- Upload signed agreement and inspection form.
- Upload 5 car photos including front, back, left, right and odometer.
- Upload additional photos if found damage on the car (optional)
- Record pickup timestamp.

Data Update/Delete:

- Update booking status to “Picked Up”.

Data Query

- Retrieve booking details.
- Check if before 30 minutes of scheduled pickup time.
- View upload forms and car photos

4. Car Return Process

Data Entry:

- Upload 6 car photos (front, back, left, right, odometer, car key)
- Report issues via checkbox.
- Enter customer feedback.
- Enter bank details for RM50 deposit refund.
- Record return timestamp.

Data Update/Delete:

- Update booking status to “Returned” or “Late Returned”.
- Update car status to “Maintenance” if issues reported, else update to “Available”.
- Update customer status to “Blacklisted” if the car is late returned.
- Update car mileage and fuel level.
- Mark car issues as resolved.
- Update car status back to “Available” after maintenance.
- Upload deposit refund receipt.
- Update booking status to “Completed”.
- Add loyalty stamp if booking duration > 12 hours.

Data Query:

- Retrieve booking details.
- Check expected return time.
- Check if return is late. (>30 minutes of the scheduled return time)
- Calculate late penalty. (RM50 per hour)
- View customer feedback and reported issues.
- View uploaded return photos.
- View cars flagged for maintenance.

5. Penalty Process

Data Entry:

- Submit penalty payment receipt by customer.

Data Update/Delete:

- Approve penalty payment by staff.
- Update customer status from “Blacklisted” to “Active”.

Data Query:

- View pending penalty payments.
- View penalty payment receipt.
- Check customer status.

6. Car Management Process

Data Entry:

- Add new car to fleet with details.
- Upload car photo.
- Set pricing rates.
- Set special features if any.

Data Update/Delete:

- Edit car details.
- Update car pricing rates.
- Change car status via dropdown. (Available, Rented, Maintenance, Inactive)

Data Query:

- View complete car list.
- Check car availability for specific dates.
- View past rental records for specific car

7. Reporting & Monitoring Process

Data Entry:

No data update or deletion are included in this process.

Data Update/Delete:

No data update or deletion are included in this process.

Data Query:

- Rental report
- Revenue report
- Booking trends graph
- Customer distribution charts by faculty and college
- Booking heatmap
- Filter analytics by time period
- Export reports to PDF/Excel
- View dashboard key metrics (total bookings, revenue, customers)

1.11 Benefit & Summary of Proposed System

The proposed system can bring several benefits by changing the current manual workflow to an automated and database-driven solution. First, it avoids redundant data, because customer, vehicle, booking, and payment information are only stored once and reused for multiple processes. It also prevents data from storing in different places by introducing a centralized database, making it easy for data retrieval. This minimizes repetitive data entry and reduces the human errors in the booking process workflow.

Besides, the centralized database also could allow real-time data access and updates. So, the staff can view the latest car availability, booking status, payment verification status and return progress instantly. Hence, staff does not need to do cross-checking to ensure all the data is up to date, and worrying the data queried is not the latest data. This improves the efficiency in staff workload and prevents issues like double book and long booking confirmation period.

Furthermore, the proposed system also provides structured relationships between entities and role-based access control to solve current data integrity and security problems in the current state. For example, sensitive information such as customer personal information, licenses and bank information is stored in a database and only can be accessed by authorized users by system boundary and managing major user views. The data security is improved as the database is backed up regularly, which is something that is lacking in the current system.

Other than that, the proposed database-driven solution also could save the time spent on data queries via manual processes. With the new solution, staff and customers can easily query data by simple user interface, making the process easy to use and improve efficiency. Additionally, it also makes the queries and retrieval of historical data easy, without needing to search back the huge amount of hardcopy records, which is very hard to be traced back. It is also very easy to do marketing analysis and reporting with the proposed solution, as the data is easily retrieved and it can be used for decision making and performance tracking. The data retrieved can even be used for machine learning dataset to be used for forecasting purposes.

To conclude, the proposed system can significantly improve service quality of Hasta Travel. It also organizes the data to store in a correct manner for data retrieval and storage purposes. Besides, it also replaces fragmented and heavily manual workflows into an integrated system that are able to support current operation mode, improve it, and make it possible for future scalability. Overall, this solution does cause a big change in the current operation mode, but it is definitely a good thing and worth the change.

1.12 Summary

In this report, we had successfully analyzed the current workflow of Hasta Travel & Tours, which relies heavily on manual processes such as WhatsApp communication, handwritten records, whiteboards, Excel spreadsheets, and Wahdah staff system. The results of analysis show some huge flaws such as data redundancy, not up to date data, inconsistent records, and heavy staff workload due to repetitive data handling.

To address these challenges, we proposed a digitalize system, which changes the mode of booking from WhatsApp and various platforms to only using websites. The digitalized system also uses a centralized database to manage the data of customer, rental, booking, car and staff. With the centralized database, Hasta Travel can enjoy the structured data storage, real-time transaction processing, and role-based access for customer, staff, and admin. Besides, with defined entities, attributes, relationships, and transaction processes, the system can ensure data consistency, integrity, and security throughout the rental process.

To conclude, the proposed solution can successfully provide Hasta Travel an efficient and reliable system that reduces the workload of operation and improves customer experience. Therefore, the problems of fragmented booking flow for customers, and the heavy workload on the staff side are successfully eliminated.

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Meeting Log #1

Date: 1 November 2025

Time: 9:00 p.m.

Member Involved:

Chun Yao Ting

Evelyn Ang

Mirza As-Siddiq bin Tohari

Muhammad Aliff bin Helmi

Objective:

To identify required outputs of project phase 1 based on the guideline and divide the tasks among members.

Discussion Summary:

In the first meeting, we had discussed the background of the project and agreed on the problem statement, including the current car rental process, comparing it with existing systems, and identifying key issues in the current process. After that, we worked on gathering the AS IS requirements, including the required data and transaction needed for the database proposal. Based on the results, all members agreed on the proposed system, including the features of the proposed system, creating the sitemap and flowchart to represent the TO BE process. Finally, we worked on the supporting analysis, including the objectives, feasibility study, persona, use case diagram, and system boundary with major user views.

	Task Distributed	Person-In-Charge
1.	Background Study	• Chun Yao Ting • Evelyn Ang
2.	Current System Workflow	• Mirza As-Siddiq bin Tohari • Muhammad Aliff bin Helmi
3.	Problem Statement	• Chun Yao Ting • Evelyn Ang
4.	Comparison with Existing System	• Mirza As-Siddiq bin Tohari • Muhammad Aliff bin Helmi
5.	Issues or Problems with Existing System	• All members
6.	AS-IS Data Requirement	• Mirza As-Siddiq bin Tohari • Chun Yao Ting
7.	AS-IS Transaction Requirement	• Muhammad Aliff bin Helmi • Evelyn Ang
8.	Proposed System	• All members
9.	Proposed System Sitemap	• Mirza As-Siddiq bin Tohari • Muhammad Aliff bin Helmi
10.	Proposed System Flowchart	• Chun Yao Ting • Evelyn Ang
11.	Objectives and Feasibility Studies	• Muhammad Aliff bin Helmi
12.	Persona	• Chun Yao Ting
13.	Use Case Diagram	• Mirza As-Siddiq bin Tohari
14.	System Boundary and Major User View	• Evelyn Ang

Meeting Log #2

Date: 28 November 2025

Time: 9:00 p.m.

Member Involved:

Chun Yao Ting

Evelyn Ang

Mirza As-Siddiq bin Tohari

Muhammad Aliff bin Helmi

Objective:

To identify remaining tasks in phase 1 and allocate the tasks among members.

Discussion Summary:

We had discussed the role assignment required for the project documentation, and we finally completed the Human Resource section by identifying the team responsibilities. Next, we planned the project timeline by creating the Work Breakdown Structure (WBS) of the project and mapping it into a Gantt chart for the project duration. In this section, we had also reviewed the database requirements and integrated them into the project documentation to ensure that the database scope of the project meets the key processes of the system. Finally, we discussed the benefits of the proposed solution in the project documentation and summarized them in order to justify the system's efficiency in managing the rental process compared to the current process.

	Task Distributed	Person-In-Charge
1.	Human Resource	• Mirza As-Siddiq bin Tohari
2.	Work Breakdown Structure & Gantt Chart	• Muhammad Aliff bin Helmi
3.	Database Requirement	• Evelyn Ang
4.	Benefit of Proposed Solution	• Chun Yao Ting

